Mid Term Examination, Autumn 2024
Course Title: Coordinate Geometry and Vector Analysis
Course Code: MAT 261

University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science & Engineering

Program: B.Sc. in CSE

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Marks: 20 **Time: 1*(one)* hour**

(Answer all questions)

Q.No.		Marks
1a	Define Polar Coordinate System.	2
1b	Convert the equations to the opposite coordinates: a) $(x^2 + y^2)^2 = 2a^2xy$ b) $r(cos3\theta + sin3\theta) = 5ksin\theta.cos\theta$	3
1c	Compute the value of K so that the following equations may represent pairs of straight lines. Also compute the intersecting points and the angle between them: $Kxy - 8x + 9y - 12 = 0.$	5
2a	Test the nature of the conics of the following equation: $34x^2 + 24xy + 41y^2 + 48x + 14y - 108 = 0$	3
2b	i) By converting to parallel axes through a property chosen point (h,k) prove that the equation $12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$ can be reduced to one containing only the terms of 2^{nd} degree. ii) Estimate the equation of parabola after rotating of axes through 30°: $x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$	7